

ALI MANTACHE

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Portfolio: <https://amgit21.github.io/portfolio>

PROFESSIONAL SUMMARY

AI engineer and full-stack developer with 5+ years delivering software and 2+ years shipping production AI systems. Builds secure REST APIs, LLM-powered workflows, and scalable web platforms (Python/FastAPI, Node.js/NestJS, React/Next.js, PostgreSQL, Docker, CI/CD). Turns R&D and product requirements into tested, reliable systems through API design, event-driven architecture, and cross-functional collaboration.

CORE SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, PHP
- **Backend:** FastAPI, NestJS, Express.js, Node.js, REST APIs, JWT, Microservices, Event-Driven Architecture
- **Frontend:** React.js, Next.js, WebSockets
- **Databases:** PostgreSQL, MongoDB, MySQL, SQL Server, Redis
- **AI & Automation:** OpenAI APIs, LangChain, Hugging Face, LoRA, Ollama, Groq, n8n, Prompt Evaluation
- **Data & Computer Vision:** Apache Kafka, Apache Spark, Delta Lake, YOLOv8, OpenCV, Roboflow
- **DevOps & Tools:** Docker, Docker Compose, GitHub Actions, CI/CD, Git, pytest, Pydantic
- **Other:** Hyperledger Besu, Agile/Scrum

PROFESSIONAL EXPERIENCE

Research and Development AI Engineer | Intelligencia.fr | Remote | 2024 - Present

Tech: Python, FastAPI, NestJS, Next.js, React, Node.js, Go, Kotlin, Android, PostgreSQL, SQL Server, Redis, Docker Compose, Hyperledger Besu, OpenAI API

- Architected core modules for an EU-funded R&D identity platform covering on-device face authentication, an Android holder app (Kotlin), and permissioned blockchain credential issuance
- Implemented Dockerized backend services (Go) for enrollment, eligibility checks, and zero-knowledge network authentication while keeping raw biometrics on-device
- Developed production FastAPI services with authentication, secure third-party integrations, and LLM-powered features
- Designed a catalog-first data platform (FastAPI, NestJS, Next.js, PostgreSQL JSONB) with adapters for SQL Server and other SQL sources, files, and external REST APIs
- Built an autonomous prompt-evaluation pipeline that filled ~50 prompt templates from a dataset, called OpenAI dynamically, scored outputs against ground truth (precision, recall, accuracy), and generated comparison charts to select the best prompt style for the use case
- Implemented CI/CD pipelines and containerized deployments with automated testing
- Authored technical documentation for architecture, security, demos, and operational runbooks

Lead Instructor, MERN Stack Bootcamps | ESA Coding Lab, ESA Business School | 2021 - 2024 (concurrent with development roles)

Tech: MongoDB, Express.js, React.js, Node.js, Python, Git, REST APIs

- Delivered 9 MERN bootcamps: 6 with ESIEE-IT/France via AFD-funded ESA Coding Lab; 3 with Tatweer Baladna (U.S. Embassy-funded, hybrid cohorts)
- Trained 300+ learners across universities and institutions on full-stack development, Python, REST APIs, Git, debugging, code review, and Agile practices

Full-Stack Developer / Web Development Lead | E-VOLVE Marketing Solutions | 2020 - 2022 (concurrent with instruction from 2021)

Tech: PHP, JavaScript, WordPress, MySQL, HTML, CSS

- Led 10+ developers delivering client WordPress sites end-to-end with SEO-oriented architecture
- Owned the full lifecycle from requirements gathering through deployment

SELECTED PROJECTS

AI Triage System - Agentic B2B Support Intake | 2026

Tech: Python, FastAPI, Groq (Llama 3.3), pytest, n8n, Gmail API, Google Sheets

- Built an AI support-intake system for classification, routing, and enrichment using FastAPI, n8n, Gmail, and LLMs
- Developed dual ingestion (concurrent batch with retries + Gmail/webhook n8n flows) producing structured JSON outputs
- Added schema validation, deterministic routing, regression tests, and Google Sheets fallback logging for failed runs

AI Chatbot - Natural Language to Dynamic Database CRUD | 2025

Tech: Python, FastAPI, PostgreSQL, Next.js, Hugging Face, LoRA, LangChain, Ollama

- Built a natural-language-to-SQL CRUD system with JWT authentication and query sanitization
- Fine-tuned TinyLlama with LoRA for CPU-efficient local inference
- Delivered a Next.js analytics UI with filter, search, sort, and CSV export

Intelligent Real-Time Data Streaming Platform | 2024

Tech: Apache Kafka, Apache Spark, Delta Lake, FastAPI, React.js, WebSockets, YOLOv8, OpenCV, Roboflow, Autodistill

- Architected a computer-vision streaming pipeline (Kafka + Spark Structured Streaming) for concurrent RTSP detection, with Delta Lake as the inference and alert sink
- Trained YOLOv8 for target industrial objects via manual Roboflow labeling and dynamic ontology-based labeling with Autodistill/zero-shot models (e.g., Grounding DINO)
- Designed microservice-per-source inputs (camera, web scraping, dummy data; IoT-ready) and tested event fusion (e.g., smoke + high temperature) to raise confirmed alerts
- Delivered FastAPI services and a React dashboard with WebSocket push alerts, sound notifications, real-time telemetry, and detection heatmaps

EDUCATION

- **Master's Degree in Computer Science and Risk Management** | Lebanese University, Faculty of Sciences | 2015 - 2017
- **Bachelor's Degree in Computer Science** | Lebanese University, Faculty of Sciences | 2012 - 2015

CERTIFICATIONS

- SE Factory (2025): DevOps, GenAI, Prompt Engineering
- ESIEE-IT France (2022): Full-Stack Web Developer
- Semicolon Academy (2022): Ethical Hacking Foundation